Charles Van Hook

cvanhook@bu.edu • (310) 613-8124 • Los Angeles, CA
linkedin.com/in/charles-van-hook • github.com/charmander542 • charlesvanhook.com

EDUCATION

Stanford University — Ph.D., Electrical Engineering *(current)* Stanford, CA
Boston University — B.S., Computer Engineering *(GPA 4.0)* May 2026

PUBLICATIONS

Fault Detection and Response for Safe Control of Artificial Muscles in Soft Robots. C. Van Hook*, R. Jing*, I. Yang, A. P. Sabelhaus. *IEEE Control Systems Letters*, vol. 9, 2025. (co-first author)

Scaling Up Parallel Decoding of LoRa Channels (LoRa-MUX). C. Van Hook, T. Adler, D. Starobinski. *IEEE ICC 2026* (IoT). (first author)

EXPERIENCE

Tesla — Robotics Engineer Palo Alto, CA | May 2026 – December 2026

- Worked on robotic finger structures for robot hands.
- Developed tactile designs for robotic sensing and contact.

Boeing — Technical Intern El Segundo, CA | 2020–2021

- Modeled helical antennas and generated radiation patterns to optimize design parameters in a team of three.
- Presented findings on helical-antenna and large-array satellite-disk efficiency; built RF strength test equipment and coordinated lab measurements with senior engineers.

Griffith Observatory — Engineering Consultant Los Angeles, CA | 2020–2023

- Created 3D models of 88 constellations and coordinated with a foundry for bronze casting of celestial-globe elements.
- Used MATLAB to evaluate mounting concepts for structural integrity and to estimate surface area / casting cost from OBJ geometry.

PROJECTS

Tiny Tapeout 8-bit Floating-Point MAC *(Verilog / ASIC)* Winter (personal)

- Designed an 8-bit floating-point multiply-and-accumulate (MAC) unit for Tiny Tapeout, targeting compact silicon for low-precision edge / ML-adjacent arithmetic.

Reimplementation of a 28nm Configurable Memory *(Cadence Virtuoso / SPICE)*

- Complete VLSI design flow: schematic capture and layout in Cadence Virtuoso through post-layout simulation.
- Verification across process corners (FF, SS, TT) with SPICE for power, performance, and area (PPA).

FPGA 5-Stage MIPS-like Pipeline *(Verilog / Xilinx)*

- Implemented a five-stage pipelined MIPS-like CPU in Verilog on Xilinx FPGA, covering fetch/decode/execute/memory/writeback and hazard handling.

ROS/SLAM Autonomous Robot *(ROS / Jetson)*

- Built a LiDAR-equipped robot with SLAM and waypoint navigation for autonomous mapping on Nvidia Jetson.

ROS/MoveIt Robotic Arm & Hexapod *(ROS / PCB / CV)*

- 3D-printed multi-DOF arm with MoveIt; hexapod with custom PCBs for 18 leg servos plus arm, IK, and vision for object pickup.

SKILLS

Languages: C++, Python, C, Verilog, MATLAB, JavaScript/React

Systems & ML: TensorFlow, OpenCV, ROS, Linux, Xilinx, Arduino

Hardware / EDA: Cadence Virtuoso, SPICE, Altium, Fusion 360, Onshape, SolidWorks

AWARDS

- 1st Place, Hardware Hackathon — October 2023
- 2nd Place, BU Lutron Competition — 2024